BRAC UNIVERSITY

Principles of Physics-II (PHY-112)

Department of Mathematics and Natural Sciences

Quiz: 02, **Section**: 02

Date: March 15, 2025

Duration: 35 Minutes | **Spring 2025 (10F-29C)** | Marks: 15

Use SI Units only. Partial Marks will be given for partially correct answers ONLY.

1. Which of the following conditions will make Gauss's Law invalid?— **(1)**

   ◯ Gaussian surfaces symmetric with $\vec{E}$ ◯ Open Gaussian surfaces ✓ ◯ Gaussian surfaces asymmetric with $\vec{E}$ ◯ Enclosed charge within the Gaussian Surface

   (Alternative) Which of the following conditions is not an absolute necessity for Gauss's Law to be valid?—

   ◯ Gaussian surfaces symmetric with $\vec{E}$ ◯ Open Gaussian surfaces ✓ ◯ Gaussian surfaces asymmetric with $\vec{E}$ ◯ Enclosed charge within the Gaussian Surface

2. The E vs r graph (for increasing r) drawn for the inside of a spherical charge distribution will have— **(1)**

   ◯ a positive slope ✓ ◯ a negative slope ◯ zero slope ◯ exponentially decreasing curve

   (Alternative) The E vs r graph (for increasing r) drawn for the outside of a spherical charge distribution will have—

   ◯ a positive slope ◯ a negative slope ◯ zero slope ◯ exponentially decreasing curve ✓

3. The integral version of Gauss's Law measures which of the following? **(1)**

   ◯ divergence of $\vec{E}$ ◯ charges not enclosed within a Gaussian surface ◯ curl of $\vec{E}$ ◯ flux of $\vec{E}$ ✓

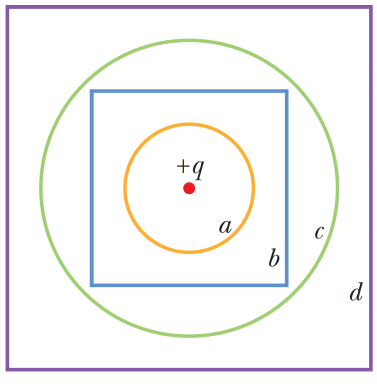
4. Which of the following things can Gauss's Law check but Coulomb's law cannot?— **(1)**

   ◯ measuring the field of a line charge ◯ measuring the flux of a volume charge distribution ◯ measuring the field inside a charge distribution ✓ ◯ measuring the field for any continuous distribution

5. The net flux of a dipole (equal and opposite charge separated by a distance) through a Gaussian surface enclosing the system will be— **(1)**

   ◯ positive ◯ negative ◯ zero ✓ ◯ the sign will depend on the shape of the surface

6. These are two-dimensional cross-sections of two Gaussian spheres and two Gaussian cubes centered on a positively charged particle. (i) Rank the net flux through the four Gaussian surfaces, smallest first. (ii) Rank the magnitudes of the electric fields on the surfaces, largest first or (Alternative) smallest first, and indicate whether the magnitudes are uniform or variable along each surface. **(3)**



(i) Since $Q_{\text{enc}} = q$ for all Gaussian surfaces, $\Phi_A = \Phi_B = \Phi_C = \Phi_D$

(ii) Since $E \propto \frac{1}{r^2}$, $E_a > E_b > E_c > E_d$. Largest to smallest. $E_d > E_c > E_b > E_a$. Smallest to largest (Alternative).

Also, spheres at a and c are symmetric with the source field. Hence, the field magnitude will be uniform along the spherical surfaces. The cubes, however, are not symmetric with the source field. Hence, the field magnitude will be variable along the cubic surfaces.

7. A flat sheet of paper of area $0.320\,\mathrm{m}^2$ is oriented so that the normal to the sheet is at an angle of 65° to a uniform electric field of magnitude $12\,\mathrm{N\,C^{-1}}$. (i) Find the magnitude of the electric flux through the sheet. (ii) For what angle θ between the normal to the sheet and the electric field is the magnitude of the flux through the sheet largest and smallest? **(3)**

(i)

$$\begin{aligned}\Phi_E &= EA\cos\theta \\ &= 12\,\mathrm{N\,C^{-1}}\cdot 0.320\,\mathrm{m}^2\cdot\cos(65^\circ) \\ &= 1.623\,\mathrm{N\,C^{-1}\,m^2}.\end{aligned}$$

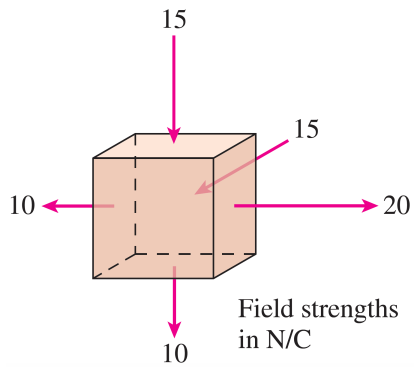
(ii) Φ_E is largest when $\cos\theta = 1$. This claims $\theta = 0^\circ$. Φ_E is smallest when $\cos\theta = 0$. This claims $\theta = 90^\circ$ or 270°.
(Alternative) A thin sheet of metal (positively charged) of area $0.500\,\mathrm{m}^2$ is oriented parallel to another thin metal sheet (negatively charged) of similar dimension, 0.5 mm apart. Both sheets are charged with 5 nC magnitude. (i) Find the strength of the electric field intensity in between the sheets. (ii) What amount of charge will double the field strength you got in (i)? [**Hint**: Start by measuring σ.]
(i) Since this is a case of two parallel plate with equal surface charge densities, the net electric field in between the plates can be found using:

$$\begin{aligned}E &= \frac{\sigma}{\epsilon_0} = \frac{Q}{\epsilon_0 A} \\ &= \frac{5\times 10^{-9}\,\mathrm{C}}{\epsilon_0} \\ &= 564.717\,\mathrm{N\,C^{-1}}.\end{aligned}$$

(ii) In order to double this field strength, the charge required on the plates can be found by rearranging the above expression:

$$\begin{aligned}E_{\text{new}} &= \frac{Q_{\text{new}}}{\epsilon_0 A} = 2E \\ Q_{\text{new}} &= 2E\epsilon_0 A \\ &= 2\cdot 564.717\,\mathrm{N\,C^{-1}}\cdot 8.854\times 10^{-12}\,\mathrm{F\,m^{-1}}\cdot 0.500\,\mathrm{m}^2. \\ &= 2.2135\times 10^{-10}\,\mathrm{C}.\end{aligned}$$

8. The cube contains no net charge. The electric field (directions shown by the vectors) is constant over each face of the cube. Does the missing electric field vector on the **front face** point in or out? What is the field strength? Your answer should be a scalar in $\mathrm{N\,C^{-1}}$ unit. **(2)**



No net charge inside means no flux is measured. All field line contributions must, therefore, be canceled.

$$x + \underbrace{-15-15}_{\text{incoming}} + \underbrace{10+20+10}_{\text{outgoing}} = 0$$

$$x = -10$$

The field strength is $10\,\mathrm{N\,C^{-1}}$, and the field vector on the front face points inward.

9. The electric field 0.5 m from a very long uniformly charged (negative) line is $1000\,\mathrm{N\,C^{-1}}$. How much charge (magnitude and sign) is contained in a section of the line of length 2.50 cm? **(2)**

Charge enclosed within a section (of length L) of a uniformly charged long line can be found using the definition of line charge density:

$$\begin{aligned} Q_{\text{enc}} &= \lambda L \\ &= E \cdot 2\pi\epsilon_0 r L \\ &= 1000\,\mathrm{N\,C^{-1}} \cdot 2 \cdot 3.1416 \cdot 8.854 \times 10^{-12}\,\mathrm{F\,m^{-1}} \cdot 0.6\,\mathrm{m} \cdot 2.50 \times 10^{-2}\,\mathrm{m} \\ &= 6.954 \times 10^{-10}\,\mathrm{C}. \end{aligned}$$